Classified into various groups

- Types of energy required to shape the material
  - Mechanical
  - Thermal
  - Electro thermal
  - Chemical and electro chemical

- Basic mechanism involve in the process
  - Erosion
  - Ionic dissolution
  - Vaporization

Classification(cont..):

- Source of energy required for material removal
  - Hydrostatic pressure
  - High current density
  - High voltage
  - Ionized material

- Medium for transfer of these energy
  - High velocity particles
  - Electrolyte
  - Electron
  - Hot gases

Classification(cont..)

- In thermal and electro thermal method- heat energy is concentrated on a small area of the work piece, to melt and vaporize the tiny bits of work material. (EDM, LBM, PAM, EBM, IBM)

- In chemical and electro chemical method- the work piece material in contact with a chemical solution is etched in a controlled manner. (ECM, ECG, ECH and ECD)

- In mechanical method- the material is removed by mechanical erosion of the work piece material. (USM, AJM, and WJM)

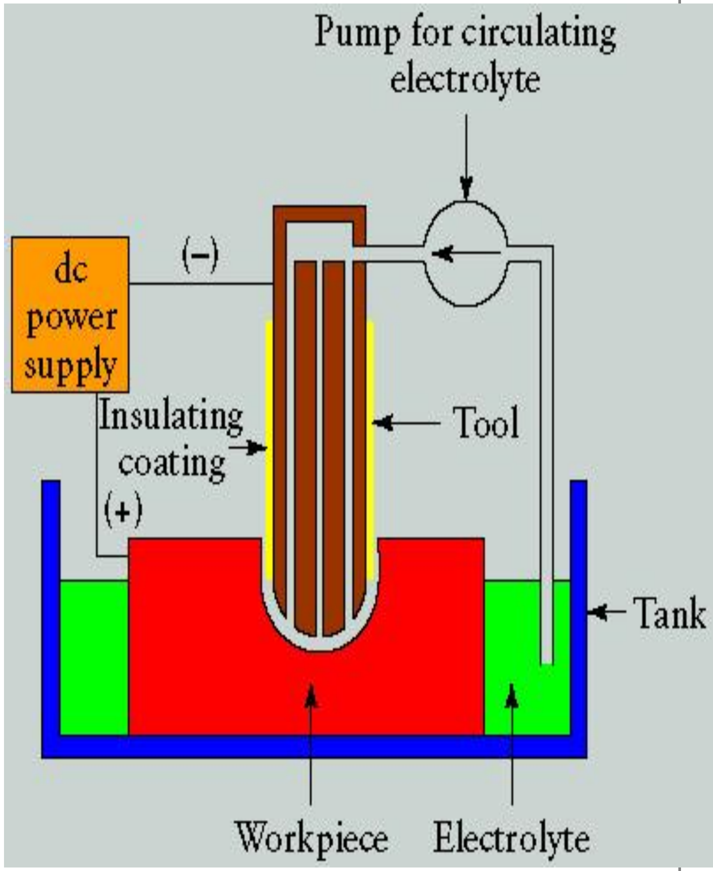
Our concern:

1. **Electro Discharge Machining (EDM)**
2. **Electro Chemical Machining (ECM)**
3. **Laser Beam Machining (LBM)**
4. **Ultrasonic Machining (USM)**
5. **Plasma Arc Machining (PAM)**

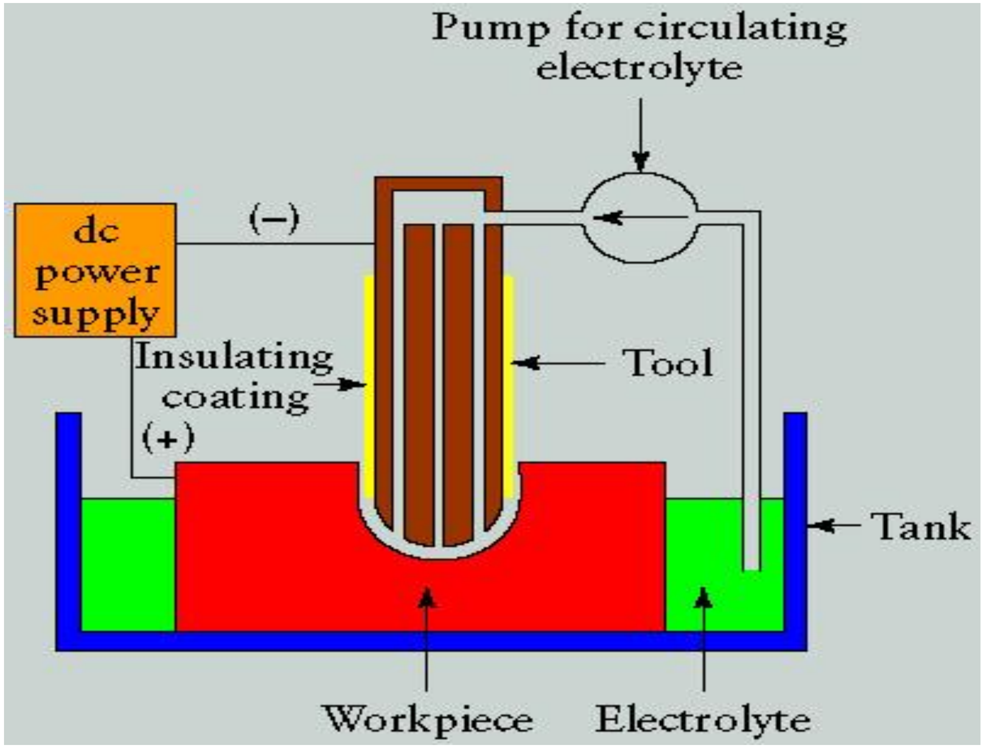
Electrochemical Machining (E.C.M.):

•Electrochemical machining (ECM) is an electrolytic material removal process involving a negatively charged shaped electrode (cathode), a conductive fluid (electrolyte), and a conductive workpiece (anode).

•ECM is an electrolytic process and its basis is the phenomenon of electrolysis, whose laws were established by Faraday in 1833.



Electrochemical Machining (E.C.M.)(cont..)



In the ECM process, the dc power source charges the workpiece positively and charges the tool negatively. As the machine slowly brings the tool and workpiece close together, within 0.010 of an inch, the power and electrolyte flow are turned on. Electrons flow across the narrow gap from negative to positive, dissolving the workpiece into the shape as the tool advances into it. The recirculating electrolytic fluid carries away the dissolved material as a metal hydroxide.

Principle:

- Based on the principle of Faraday's law of electrolysis.

Characteristic:

- Small gap between tool and work-piece (≈ 0.5mm).
- Wok-piece stationary, feed by tool.
- Electrolyte: Aqueous solution of common salt, dilute acid.
- Pressure of electrolyte ≈ 14 kg/cm2
- Velocity of electrolyte = 30 to 60 m/sec.
- Temperature of electrolyte = 25 to60°C .
- Voltage of electrolyte = 5 to 15V

ECM: Advantages

- Components are not subject to either thermal or mechanical stress.
- There is no tool wear in ECM.
- Non-rigid and open work pieces can be machined easily as there is no contact between the tool and work piece.
- Complex geometrical shapes can be machined repeatedly and accurately.
- ECM is a time saving process when compared with conventional machining
- During drilling, deep holes can be made or several holes at once.
- Fragile parts which cannot take more loads and also brittle material which tend to develop cracks during machining can be machined easily in ECM
- Surface finishes of 25 µ in. can be achieved in ECM

ECM: Disadvantages

- Keeping the solution conductivity constant.
- More expensive than conventional machining.
- Need more area for installation.
- Electrolytes may destroy the equipment.
- Not environmentally friendly (sludge and other waste)
- High energy consumption.
- Chemical attack by electrolytes.
- The danger of a burn in the case of a short circuit between the positive and negative leads.
- The danger of a fire damp explosion.
- Material has to be electrically conductive

ECM Applications

•The most common application of ECM is high accuracy duplication. Because there is no tool wear, it can be used repeatedly with a high degree of accuracy.

• It is also used to make cavities and holes in various products.

• It is commonly used on thin walled, easily deformable and brittle material because they would probably develop cracks with conventional machining.

Products

•The two most common products of ECM are turbine/compressor blades and rifle barrels. Each of those parts require machining of extremely hard metals with certain mechanical specifications

•Some of these mechanical characteristics achieved by ECM are:

*Stress free grooves.
*Any groove geometry.
*Any conductive metal can be machined.
*Repeatable accuracy of 0.0005”.
*High surface finish.
*Fast cycle time.

Function of Electrolyte in ECM:

•Completes the electric circuit between tool and work piece.
•Allows desirable machining to occur.
•Carries away products of reaction from the zone of machining.
•Carries away heat generated during chemical reactions.

Properties of Electrolyte For ECM:

•High electrical conductivity.
•Chemical stability.
•High specific heat and low viscosity.

Electrical Discharge Machining (EDM):

•Its a manufacturing process whereby a desired shape is obtained using electrical discharges (sparks).

•Material is removed from the workpiece by a series of rapidly recurring current discharges between two electrodes, separated by a dielectric liquid and subject to an electric voltage.

•One of the electrodes – ‘tool-electrode’ or ‘tool’ or ‘electrode’.

•Other electrode - workpiece-electrode or ‘workpiece’.

•As distance between the two electrodes is reduced, the current intensity becomes greater than the strength of the dielectric (at least in some points) causing it to break.

Electrical Discharge Machining (EDM)(cont..):

- It is a process of metal removal based on the principle of material removal by an interrupted electric spark discharge between the electrode tool and the work piece.

- In EDM, a potential difference is applied between the tool and workpiece.

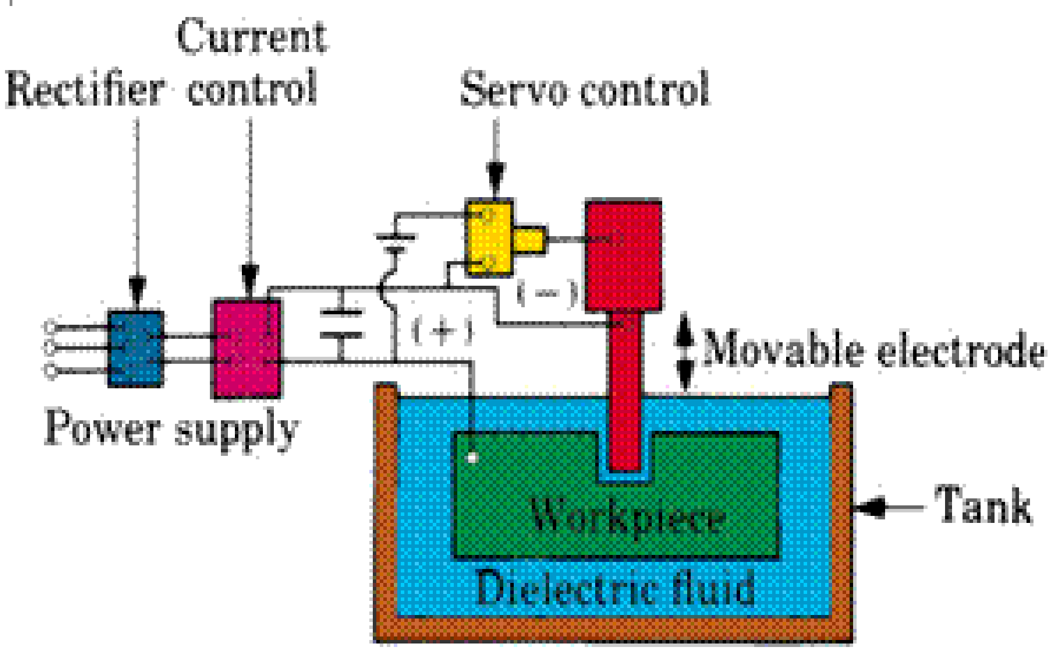
- Essential - Both tool and work material are to be conductors.

- The tool and work material are immersed in a dielectric medium.

- Generally kerosene or deionised water is used as the dielectric medium.

- A gap is maintained between the tool and the workpiece

Electrical Discharge Machining (EDM)(cont..):



•Material removal occurs due to instant vaporization of the material as well as due to melting.
•The molten metal is not removed completely but only partially.

Applications:

- Drilling of micro-holes, thread cutting, helical profile milling, rotary forming, and curved hole drilling.
- Delicate work piece like copper parts can be produced by EDM.
- Can be applied to all electrically conducting metals and alloys irrespective of their melting points, hardness, toughness, or brittleness.
- Other applications: deep, small-dia holes using tungsten wire as tool, narrow slots, cooling holes in super alloy turbine blades, and various intricate shapes.
- EDM can be economically employed for extremely hardened work piece.
- Since there is no mechanical stress present (no physical contact), fragile and slender work places can be machined without distortion.
- Hard and corrosion resistant surfaces, essentially needed for die making, can be developed.

Principle:

When a potential difference is applied between two conductors immersed in a dielectric fluid. The fluid will be ionized and a spark will occurs. If the potential difference is maintained, the spark will develop into an arc.

Characteristic of EDM:

- A suitable gap between tool and work-piece called spark gap must be maintained (0.01 to 0.5 mm)
- More erosion on anode (+ve), so the work-piece is kept +ve.
- Dielectric fluid (Paraffin, kerosene, transformer oil etc.)
- D.C. voltage: 40-300V,Current: 0.5to400Amp, local temperature 10000°C
- A true replica of the tool surface will be produced on the work-piece.
- Dielectric fluid is circulated to maintain a certain temperature.

- **Function of Dielectric Fluid:**
  - Serves as a conducting medium and convey the spark.
  - Cools the work-piece and tool.
  - Carries away the eroded metal.

- **Properties of Dielectric Fluid:**
  - Should not evolve toxic vapours or gases.
  - Must be inflammable.
  - Chemically inert on tool and work-piece

- **Tool material:**
  - Brass, copper or alloy of copper, cast iron used as cutting tool material.

Ultrasonic Machining (USM):

•In the process of Ultrasonic Machining, material is removed by micro-chipping or erosion with abrasive particles.

•In USM process, the tool, made of softer material than that of the workpiece, is oscillated by the Booster and Sonotrode at a frequency of about 20 kHz with an amplitude of about 25.4 um (0.001 in).

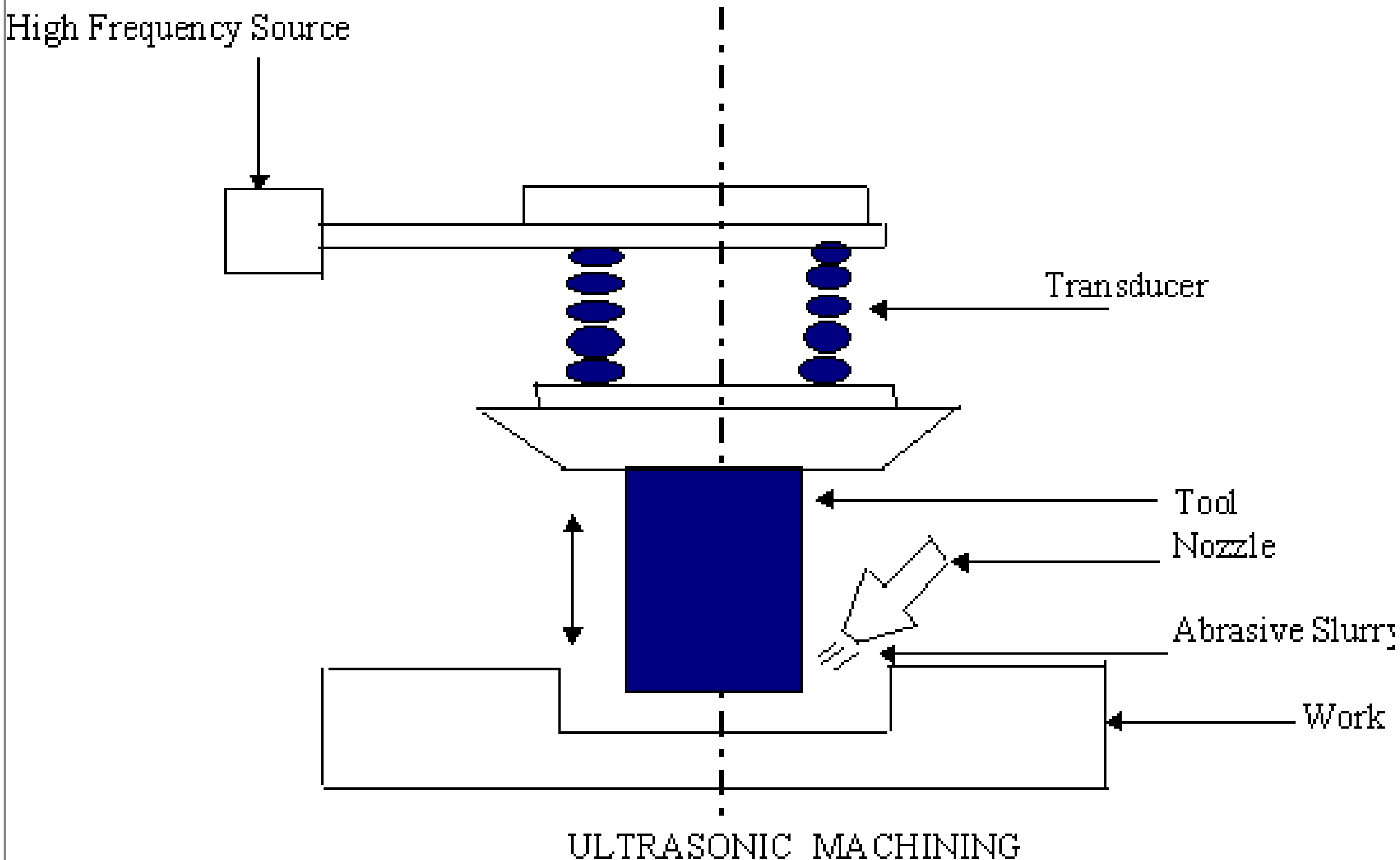
• The tool forces the abrasive grits, in the gap between the tool and the workpiece, to impact normally and successively on the work surface, thereby machining the work surface.

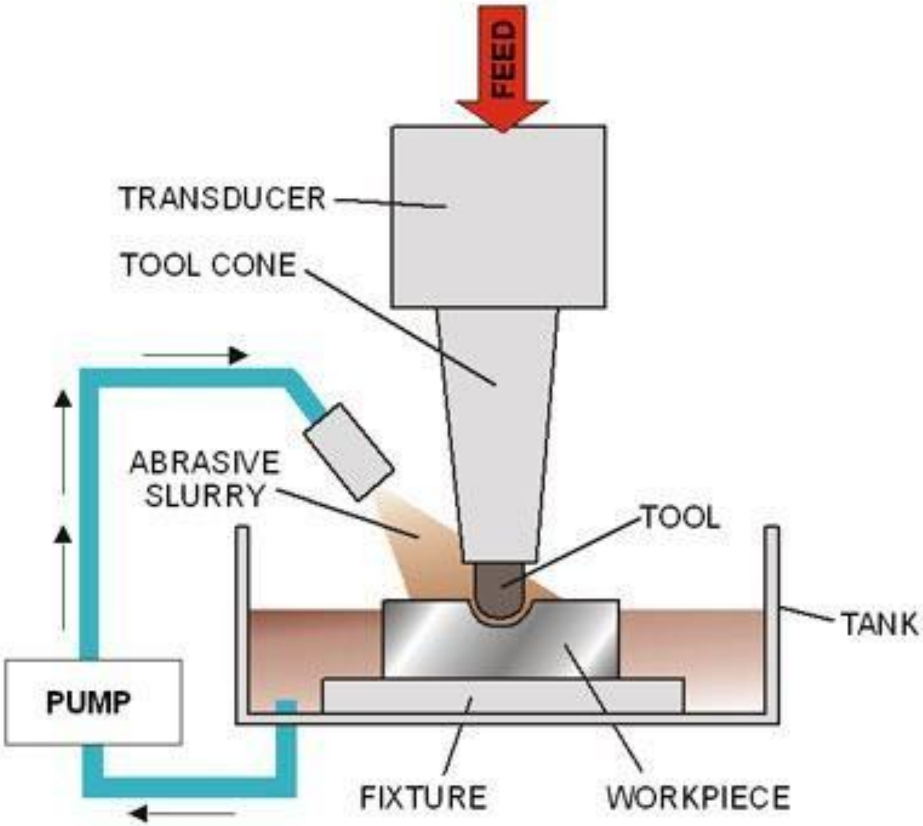
•During one strike, the tool moves down from its most upper remote position with a starting speed at zero, then it speeds up to finally reach the maximum speed at the mean position.

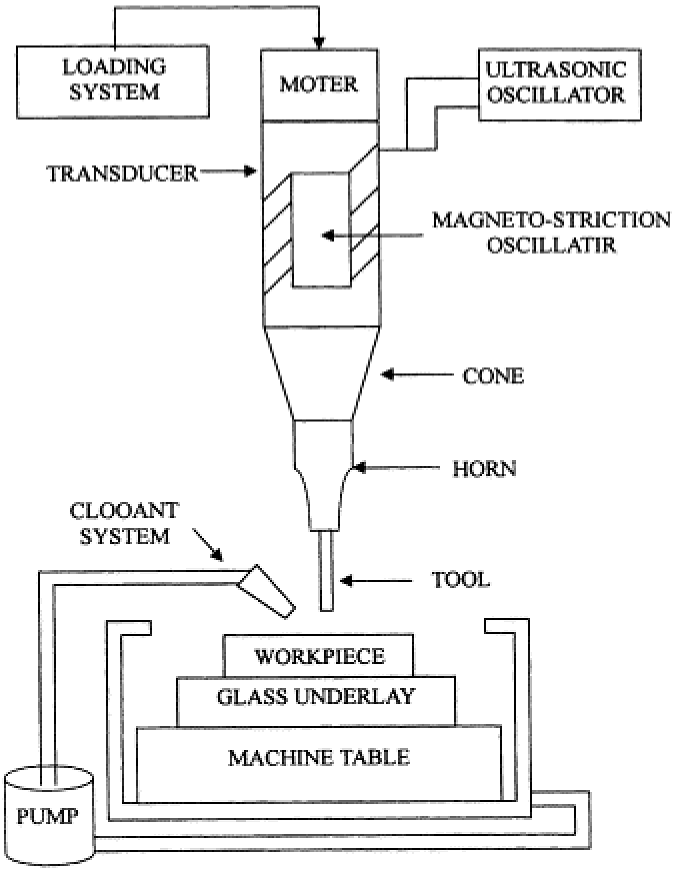
•Then the tool slows down its speed and eventually reaches zero again at the lowest position.

•When the grit size is close to the mean position, the tool hits the grit with its full speed

Ultrasonic Machining (USM) (cont..)




FEED
TRANSDUCER
TOOL CONE
ABRASIVE SLURRY
TOOL
TANK
PUMP
FIXTURE
WORKPIECE


LOADING SYSTEM
MOTER
ULTRASONIC OSCILLATOR
TRANSDUCER
MAGNETO-STRICTION OSCILLATIR
CONE
HORN
CLOOANT SYSTEM
TOOL
WORKPIECE
GLASS UNDERLAY
MACHINE TABLE
PUMP

Ultrasonic Machining (USM) (cont..)

- The smaller the grit size, the lesser the momentum it receives from the tool.

- Therefore, there is an effective speed zone for the tool and, correspondingly there is an effective size range for the grits.

- In the machining process, the tool, at some point, impacts on the largest grits, which are forced into the tool and work piece.

- As the tool continues to move downwards, the force acting on these grits increases rapidly, therefore some of the grits may be fractured.

- As the tool moves further down, more grits with smaller sizes come in contact with the tool, the force acting on each grit becomes less.

- Eventually, the tool comes to the end of its strike, the number of grits under impact force from both the tool and the workpiece becomes maximum.
- Grits with size larger than the minimum gap will penetrate into the tool and work surface to different extents according to their diameters and the hardness of both surfaces

Advantages ultrasonic machining:

- Machining of any material regardless of conductivity.
- Precision machining of brittle hard materials.
- Does not produce electric, thermal or chemical defects at the surface.
- Can drill circular or non-circular holes in very hard materials.
- Less stress because of its non-thermal nature.

Disadvantages ultrasonic machining:

- Low material removal rate.
- Tool wears fast.
- Machining area and depth are quite restricted.

❖ **Abrasives:**

Selection on the type of work-piece material, hardness of work- piece, metal removal rate, desired surface finish etc.

❖ **Types:**

Al_2O_3, Sic, Boron Carbide etc.

Water slurry with 30 to 60% by volume of abrasive.

❖ **Applications:**

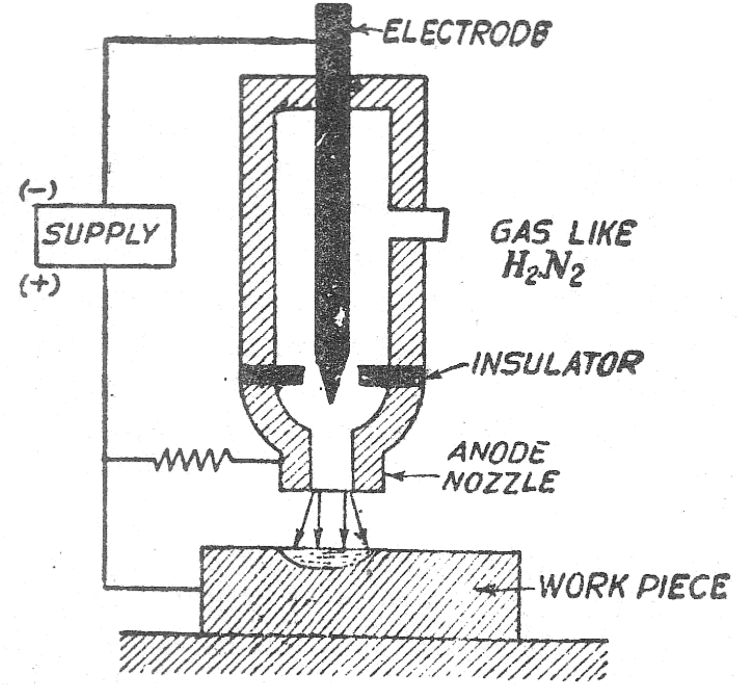
1) For machining intricate shaped products.
2) Small holes.
3) Machining of hard and non-conductive material.

Plasma Arc Machining(PAM):

- When a flowing gas is heated to sufficiently high temperature of the order of 16500°C to become partially ionized, it is known as "Plasma".

- Heat for machining produced by high temperature plasma and also due to direct electro bombardment.

- Metal removal rate can be increased by increasing gas flow rate .

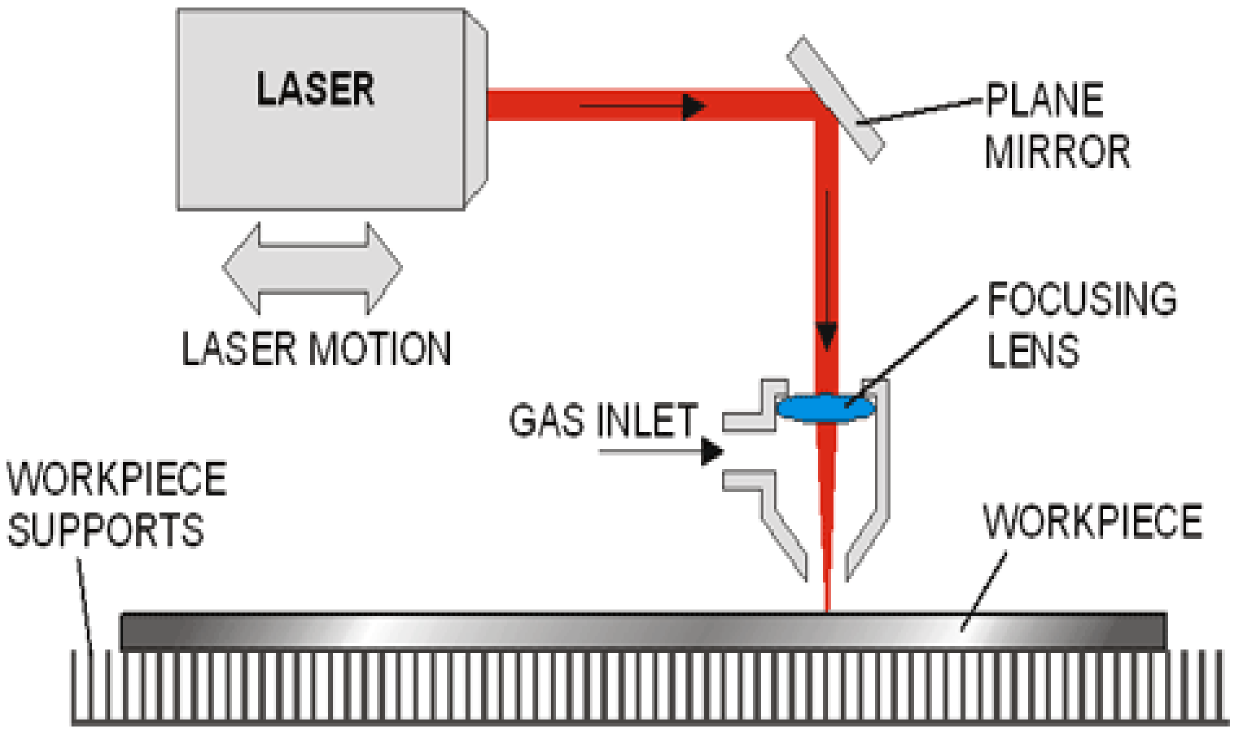


There are two type of plasma arc system

1. Transferred arc type
2. Non transferred are type

Laser Beam Machining (LBM):

- Laser is an electro magnetic radiation.
- It produce a monochromatic light called laser beam and is focused on the work-piece by lense to give extremely high temperature to melt and vaporized any material.
- For production of laser beam, generally ruby rod used in which aluminum is the main ingredient
- Laser beam techniques involves
  - Pumping of energy.
  - Production of stimulated effect


LASER
PLANE MIRROR
LASER MOTION
FOCUSING LENS
GAS INLET
WORKPIECE SUPPORTS
WORKPIECE

Advantages:

- No direct contact between tool and work-piece
- Laser beam can be sent from longer distance
- Micro shaped hole can be laser drill.

Limitation:

- High initial cost
- Overall low efficiency
- Not able to drill too deep hole

❖ **Applications:**

1) Small holes, suitable for fine and minute holes.

2) Welding like non-conducting and refractory (high temperature protect) material.

3) Cutting complex profiles on hard materials.

4) Partial cutting etc.